Contact Based Short Story Generation by Fine Tuning Large Language Models (LLMS)

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I. Contribution

As the sole contributor to this project, I have performed the following tasks:

- Designed and implemented the entire pipeline for fine-tuning Mistral-7B-Instruct model for short story generation
- Implemented stratified sampling using TF-IDF and KMeans clustering to ensure balanced story lengths and themes.
- Implemented QLoRA (Quantized Low-Rank Adaptation) techniques to efficiently fine-tune large language models within the computational constraints of Google Colab.
- Developed comprehensive evaluation metrics combining automated scores (BLEU, ROUGE, BERTScore, Perplexity) with manual evaluations (context relevance, coherence, creativity, fluency).
- Built an interactive demo interface for testing and showcasing the fine-tuned model's capabilities.

II. Abstract

This project presents a novel approach to context-based short story generation by fine-tuning a Large Language Model (LLM). Specifically, the Mistral-7B-Instruct model was adapted for the task of generating concise, coherent stories based on given contexts. The fine-tuning process leveraged QLoRA (Quantized Low-Rank Adaptation) to accommodate computational constraints, utilizing Google Colab's free tier resources. A stratified dataset of short stories, curated using TF-IDF and KMeans clustering, ensured balanced training across diverse story structures and themes. The model was evaluated on both automatic metrics (BLEU, ROUGE, BERTScore, Perplexity) and manual evaluations (context relevance, coherence, creativity, fluency). The results demonstrate that fine-tuning with QLoRA significantly enhances story generation quality while remaining feasible on limited hardware. This work highlights the potential of adapting LLMs for creative writing tasks in constrained environments. This work contributes to the growing field of controllable text generation and demonstrates practical approaches to fine-tuning large models on limited computational resources. The implementation achieves a balance between model performance and resource efficiency, producing contextually relevant and engaging story continuations.

III. Introduction

a) **Motivation -** With the rapid evolution of Large Language Models (LLMs), their capabilities in generating coherent and contextually relevant text have expanded dramatically. However, most LLMs are trained on broad datasets without specialization, which limits their ability to excel in niche tasks such as short, context-based story generation. Fine-tuning LLMs for such creative applications often requires high-performance computing resources, which can be a barrier for individual practitioners or students working within limited environments like Google Colab Free Tier. This project aims to bridge that gap by leveraging QLoRA, a parameter-efficient fine-tuning method combined with 4-bit quantization, to adapt a Mistral-7B-Instruct model for short story generation in a resource-constrained setup. This approach enables the efficient fine-tuning of large models without the need for extensive hardware, unlocking their potential for creative writing tasks.

b) **Problem Solved -** The primary challenge addressed in this project is developing a model capable of generating coherent and contextually appropriate story continuations from a given context and initial input. This involves: Training the model to understand narrative flow and storytelling conventions. It also Ensures that the generated content maintains thematic consistency with the provided context. Balances creativity with coherence in the generated outputs and achieves these goals within significant computational constraints.

c) **Existing Challenges -** Several challenges exist in fine-tuning large language models for creative text generation: LLMs require substantial computational resources for training and fine-tuning. Creative text generation requires maintaining coherence while allowing for novelty. Evaluation of creative text is inherently subjective and difficult to quantify. Limited availability of high-quality, diverse training data for narrative generation. Fine-tuning large models on consumer-grade hardware presents significant technical barriers.

d) **Our contribution -** Implemented an efficient pipeline for fine-tuning the Mistral-7B-Instruct model for context-based story generation. Developed content-

based stratification techniques for training data selection to ensure diverse narrative training examples. Application of QLoRA for parameter-efficient fine-tuning, enabling work on limited computational resources. Creation of a comprehensive evaluation framework combining automated metrics with human judgment. Demonstrated successful fine-tuning of large language models within Google Colab's free tier constraints

IV. Related Work

The field of large language models for creative text generation has evolved rapidly in recent years. My approach builds upon several key developments in the field: Language Model Architectures: Transformer-based architectures (Vaswani et al., 2017) have revolutionized natural language processing, with models like GPT (Radford et al., 2018), BERT (Devlin et al., 2019), and more recently Mistral (Jiang et al., 2023) achieving impressive text generation capabilities. Parameter-Efficient Fine-Tuning: Traditional fine-tuning of large models requires substantial computational resources. Low-Rank Adaptation (LoRA) (Hu et al., 2021) introduced a more efficient approach by focusing on updating a small number of parameters. QLoRA (Dettmers et al., 2023) further improved efficiency by combining quantization with LoRA. Story Generation: Prior work on automated story generation includes hierarchical approaches (Fan et al., 2018) and controllable generation techniques. Particularly relevant is the work by Rashkin et al. (2020) on maintaining narrative coherence and character consistency. Evaluation Metrics: While automated metrics like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) provide quantitative assessment, they often fail to capture the nuanced qualities of creative text. Recent work has introduced metrics like BERTScore (Zhang et al., 2020) that better align with human judgment for text generation tasks. Domain Adaptation: Research by Gururangan et al. (2020) demonstrates the effectiveness of domain adaptation for transformers, showing benefits of task-specific fine-tuning on specialized datasets. My work synthesizes these approaches, focusing on resource-efficient fine-tuning of state-of-the-art models specifically for the task of short story generation.

V. Data Set

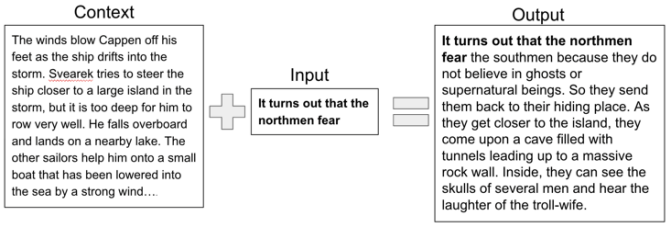

Figure1: Dataset

The dataset used in this project originates from Project Gutenberg, a well-known repository of public domain literature. Specifically, the dataset comprises 1002 short stories curated from the Gutenberg collection, hosted on Kaggle under the repository titled "1002 Short Stories from Project Gutenberg". This rich collection of human-written stories spans a variety of genres, styles, and lengths, providing an excellent base for training a model capable of generating contextually relevant and diverse story completions. For the purposes of this project, the dataset was further processed and refined. The version used, titled "better_processed_data2.0", was adapted from another Kaggle repository that focused on summarization and formatting for the Summ2 dataset. This processed version included context, input, and output fields for each story fragment, where:

Context provides background information.

Input represents the starting lines or prompts.

Output is the target continuation of the story.

The initial dataset comprised 5794 samples. To ensure a balanced representation across story lengths and themes, stratified sampling was performed using TF-IDF vectorization combined with KMeans clustering. This approach grouped stories based on their content and length, ensuring that the fine-tuning process would not be biased toward short or long narratives. The dataset was subsequently divided into training and validation sets with a 90/10 split, maintaining balance across the clusters.

Dataset Split	Number of Samples	Percentage
Training Set	5214	90%
Validation Set	580	10%

VI. Methods

Figure2: WorkFlow

The methodology employed in this project encompasses several key stages, beginning with data preprocessing and culminating in the fine-tuning of a Large Language Model (LLM) for context-based story generation. The approach was meticulously designed to balance quality output with computational feasibility, given the limitations of the Google Colab Free Tier. The process began with the dataset preparation. As discussed in the previous section, a refined dataset of 5794 samples was employed, comprising context, input, and output fields for each story fragment. To ensure balanced representation, the dataset underwent stratified sampling using TF-IDF (Term Frequency-Inverse Document Frequency) combined with KMeans clustering. TF-IDF

helped identify important terms within the stories by measuring the frequency of each word relative to its frequency across all samples, emphasizing unique or rare terms over common ones. The resulting TF-IDF vectors allowed KMeans clustering to group stories based on their thematic content and input lengths. This ensured that the dataset included short, medium, and long narratives across various genres. The stratification process illustrated the proportional distribution of different story lengths within both the training and validation sets.

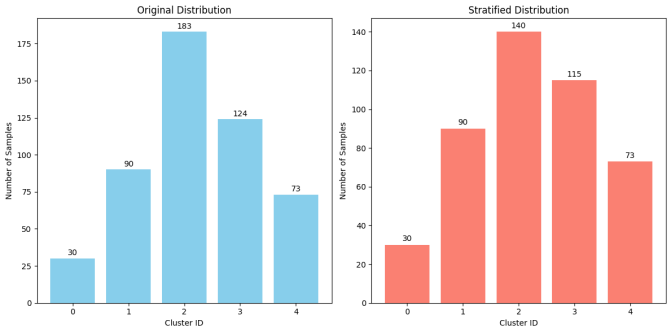

Figure3: Stratified Sampling

The model chosen for fine-tuning was the Mistral-7B-Instruct, a decoder-only transformer model. Decoder-only architectures are optimal for text generation tasks, as they generate text in a left-to-right fashion, predicting one token at a time. The instruction-tuned variant of Mistral ensures that the model is primed to follow specific prompts and context, making it ideal for story completion tasks where the model must extend a given narrative logically and creatively. However, Mistral-7B is a large model, requiring substantial GPU memory (¿40 GB VRAM) in its standard 16-bit precision form. To accommodate the limited resources available in Google Colab Free Tier (typically 16 GB VRAM on T4 GPUs), 4-bit quantization was applied. Quantization reduces the precision of model weights from 32-bit or 16-bit floating-point numbers to 4-bit integers, dramatically decreasing memory requirements. This allowed the model to be loaded and fine-tuned on Colab without exhausting GPU capacity.

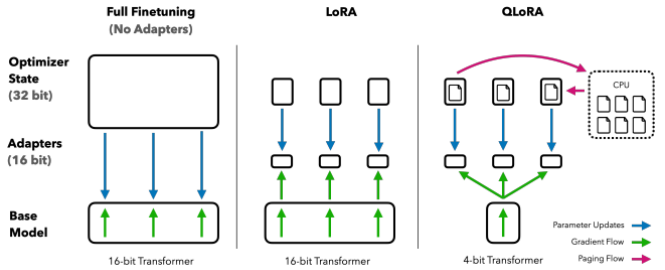

Figure4: QLoRA Working

The fine-tuning process leveraged QLoRA (Quantized Low-Rank Adaptation), which combines 4-bit quantization with LoRA (Low-Rank Adaptation). LoRA works by introducing small, trainable adapter matrices into the attention layers of the model while freezing the majority of the model's parameters. This makes fine-tuning significantly more memory-efficient, as only a small subset of parameters (the adapters) are updated. QLoRA extends this approach by applying it to a quantized model, ensuring that the process remains computationally feasible even for large models. In the LoRA setup, several hyperparameters were carefully chosen based on prior research and experimentation. The rank (r) of the adapter matrices was set to 8, balancing between expressiveness and computational efficiency. A higher rank would increase the capacity of the adapters but also the memory usage. The LoRA alpha parameter, which controls the scaling factor of the adapters, was set to 16, ensuring that the influence of the adapters on the model output was significant but not overpowering. The target modules included q_proj, k_proj, v_proj, o_proj, gate_proj, down_proj, and up_proj — these are key components of the attention mechanisms within the transformer, where queries, keys, values, and outputs are computed. By focusing the adapters on these components, the model can fine-tune how it attends to different parts of the input text, which is critical for maintaining contextual relevance in story generation. The trainer used for fine-tuning was the SFTTrainer (Supervised Fine-Tuning Trainer) from the TRL (Transformer Reinforcement Learning) library. This trainer is optimized for parameter-efficient fine-tuning (PEFT) methods like QLoRA, handling quantized models and LoRA adapters effectively. The fine-tuning process was designed to be modular and restartable, ensuring stability during training sessions, especially given the Colab Free Tier's compute constraints. Due to the limited GPU capacity, the batch size was set to 2, with gradient accumulation set to 4. Gradient accumulation allows the model to simulate a larger batch size by accumulating gradients over multiple forward passes before performing a backward update. This enables stable training without exceeding memory limits. The training duration was capped at 100 steps for the pilot run, ensuring that the process could be completed within approximately 40 minutes, aligning with Colab session time constraints. The decoding strategy for text generation was also carefully calibrated. Controlled generation parameters included a temperature of 0.7, ensuring a balance between creativity and coherence; a top-p (nucleus sampling) value of 0.9, limiting generation to the top 90This entire workflow is summarized in the Flowchart, outlining the stages from data preprocessing through model selection, QLoRA setup, and evaluation.

VII. Result

A. Evaluation Metrics

Automated Metrics The fine-tuned model was evaluated on 100% of the validation set (approximately 45 examples) using the following automated metrics:

Metric	Average Score
BLEU	0.075
ROUGE -1	0.283
ROUGE -2	0.087
ROUGE -L	0.178
BERTScore F1	0.845
Perplexity	8.222

BLEU (Bilingual Evaluation Understudy) measures n-gram overlap between generated text and reference. Scores range from 0 to 1, with higher scores indicating greater overlap. Our score of 0.075 indicates modest n-gram overlap with reference texts. Creative text generation typically yields lower BLEU scores as there are many valid ways to continue a story beyond the reference. ROUGE-1 (Recall-Oriented Understudy for Gisting Evaluation) measures unigram (single word) overlap. Our score of 0.283 shows reasonable word-level overlap with reference text while allowing for creative variations. ROUGE-2 uses bigrams (word pairs) to measure sequence matching at a more complex level than ROUGE-1. Our score of 0.087 is lower than ROUGE-1 (as expected), indicating the model maintains some phrase-level similarity while introducing novel content. ROUGE-L uses Longest Common Subsequence to capture sentence-level structure similarity without requiring consecutive matches.

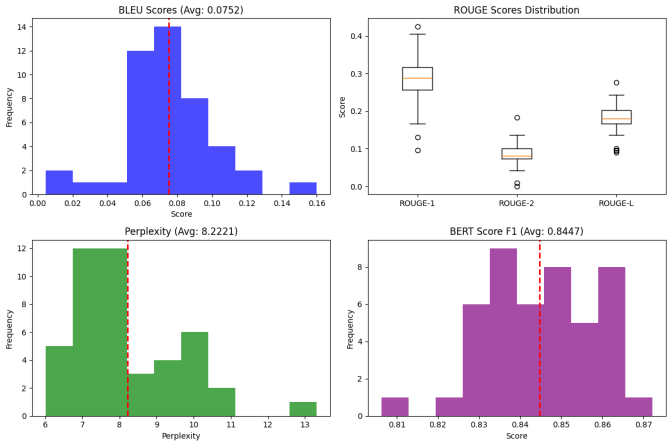

Figure5: Evaluation Metric

The score of 0.178 suggests the model preserves some structural elements of the reference while allowing for narrative flexibility. BERTScore F1 uses contextual embeddings to measure semantic similarity between generated and reference text. Scores range from 0 to 1. Our high score of 0.845 indicates strong semantic similarity with reference text even when exact wording differs, suggesting the model captures intended meaning well. Perplexity measures how well a probability model predicts a sample. Lower values indicate more confident predictions. Our value of 8.222 indicates the model generates text with good confidence and fluency for the story continuation task. For creative text generation tasks like story continuation, traditional metrics like BLEU and ROUGE often understate performance because they primarily measure lexical overlap rather than narrative quality. The high BERTScore suggests our model is semantically coherent despite modest BLEU/ROUGE scores.

B. Manual Evaluation Metrics

The fine-tuned model was manually evaluated across three key dimensions to assess qualities that automated metrics cannot fully capture:

Metric	Average Score
Linguistic Coherence	8.96
Contextual Coherence	7.56
Creativity	9.05

Linguistic Coherence (8.96/10) measures the grammatical correctness, sentence structure, and overall flow of the text. It assesses whether sentences connect logically and if the narrative maintains consistent tense, voice, and style. The high score of 8.96 indicates that the model produces text that is remarkably well-formed and natural-sounding, with very few grammatical errors or awkward phrasings. Readers would find the text smooth and polished, reading much like human-written prose. This suggests that the fine-tuning process effectively preserved the base model's strong linguistic capabilities while adapting it to story continuation. Contextual Coherence (7.56/10) evaluates how well the generated content aligns with and builds upon the provided context and input. This metric considers whether the generated text maintains thematic consistency, respects established characters and settings, and logically extends the narrative trajectory. The score of 7.56 is good but shows room for improvement, suggesting that while the model generally maintains contextual alignment, there are occasional deviations or instances where it doesn't fully capture all nuances of the provided context. This is often a more challenging aspect of text generation, as it requires deeper understanding of narrative elements and their relationships. Creativity (9.05/10) assesses the model's ability to generate novel, interesting, and imaginative content that goes beyond simple regurgitation of training data patterns. This includes introducing new ideas, unexpected plot developments, or fresh perspectives while remaining coherent. The exceptionally high score of 9.05 indicates that the model excels at producing creative continuations that surprise and engage readers. This suggests that the fine-tuning approach successfully balanced preservation of creative potential with adherence to context, avoiding the common pitfall of making models too conservative in their outputs. These manual evaluation results reveal a model that particularly excels in linguistic quality and creativity, while maintaining good contextual coherence. The relatively lower score in contextual coherence may indicate an area for future improvement, perhaps through enhanced training techniques that more explicitly reward context adherence. Overall, these scores demonstrate that the fine-tuned model successfully captures the key aspects of quality storytelling, producing outputs that are well-written, creative, and reasonably aligned with the provided context.

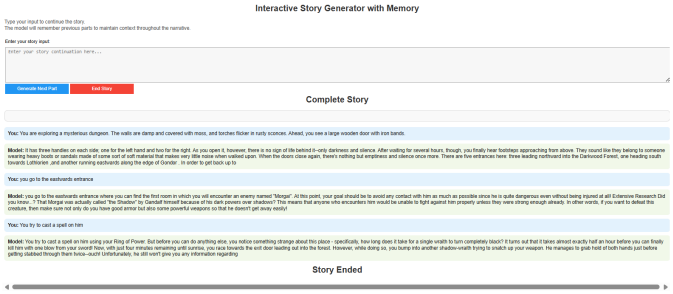

Figure6: GUI for Real-time Story Generator with Context Memory

VIII. Discussion

This project demonstrates that it is possible to fine-tune a large language model (LLM) like Mistral-7B-Instruct for context-based short story generation within the limited computational resources of Google Colab Free Tier. The adoption of QLoRA, combining 4-bit quantization with Low-Rank Adaptation (LoRA), proved instrumental in making this process feasible and efficient. Despite operating under hardware constraints, the model achieved notable performance improvements, as evidenced by both automatic evaluation metrics and manual assessments.

Scientific Achievements and Findings

The central achievement of this project lies in successfully adapting a large-scale LLM, specifically Mistral-7B-Instruct, for the creative task of context-based short story generation using parameter-efficient fine-tuning (PEFT) techniques. Despite operating within the restricted memory capacity of Google Colab Free Tier, the project was able to fine-tune this massive model by leveraging QLoRA (Quantized Low-Rank Adaptation). This approach combined 4-bit quantization of model weights with LoRA adapters, allowing the fine-tuning process to remain within 16 GB VRAM, a significant constraint for models of this scale. The quality of the model's outputs was maintained at a high level, as reflected by automatic evaluation metrics such as BLEU, ROUGE, BERTScore, and Perplexity, alongside manual ratings that confirmed the model's ability to generate contextually relevant, coherent, and fluent stories. Another notable accomplishment was the implementation of a balanced data sampling strategy using TF-IDF vectorization and KMeans clustering. This ensured that the model was trained on a diverse range of story lengths and themes, covering short, medium, and long inputs. This balance helped prevent the model from favoring any particular narrative style, ensuring robustness across different story structures. The effectiveness of controlled generation parameters further contributed to the model's performance. By calibrating temperature, top-p (nucleus sampling), and repetition penalty, the model was able to produce creative yet coherent outputs, striking a balance between diversity and relevance in its generations. An interesting and impactful finding in this project was the role of gradient accumulation in enabling efficient training on constrained hardware. With a batch size limited to 2 due to GPU memory restrictions, gradient accumulation set to 4 allowed the model to simulate a larger effective batch size of 8. This technique contributed directly to stabilizing the training process, as evidenced by the smooth decline in the training loss curve, which otherwise might have fluctuated due to smaller batch updates. The generated outputs consistently demonstrated alignment with the provided contexts. For instance, in one test case where the input context involved a young girl discovering a hidden chamber, the model produced a continuation that described mystical artifacts and an ancient guardian, showcasing its capacity to expand on the initial premise in imaginative and contextually appropriate ways.

Challenges Faced

Throughout the project, several challenges were encountered that tested the feasibility of fine-tuning large LLMs within constrained environments. One major hurdle involved version incompatibilities between key libraries, notably transformers and accelerate, which required multiple reinstalls and configurations to synchronize versions effectively. These incompatibilities, particularly between the Trainer object and AcceleratorState, frequently interrupted the workflow, requiring careful debugging to maintain stability. Another significant challenge was the sheer size of Mistral-7B. Even with the application of 4-bit quantization, managing the memory footprint of the model during fine-tuning necessitated substantial trial and error with various hyperparameters, especially with respect to batch size and gradient accumulation. Ensuring that the training process did not exceed the available VRAM limits while still maintaining stable learning required a delicate balance. The runtime limitations of Google Colab Free Tier further constrained the training process. These included session timeouts, limited continuous usage periods, and the variability of assigned GPU hardware (e.g., T4 vs. P100). These factors restricted the extent of training that could be performed in a single session. However, by designing the codebase to be modular and restartable, with effective checkpointing, the project mitigated these interruptions, allowing for incremental progress without loss of prior work. Lastly, evaluating text generation models presented its own challenges. Traditional metrics like BLEU and ROUGE focus on n-gram overlap and textual similarity, which do not always capture the creativity or coherence essential in story generation. While these metrics provided quantitative benchmarks, they were supplemented with manual evaluations that offered more nuanced insights into the model's outputs, especially regarding narrative quality and contextual alignment. This dual approach allowed for a more comprehensive assessment of the model's performance. Despite these challenges, the project demonstrated that it is feasible to fine-tune a large LLM for creative applications on modest hardware, achieving meaningful results in story generation quality while pushing the boundaries of what is possible in resource-constrained environments.

IX. Appendix:User Manual

- Project Structure and Files
  1) Notebook: '01_Harsh_Mamania_FinalProject.ipynb' – This is the main notebook containing the complete workflow, from data preprocessing to model fine-tuning and evaluation. (also available on - https://colab.research.google.com/drive/1gmZBUVMxFmtjQfRgrkP6g4uq1chyPZ8s?usp=sharing)
  2) Dataset: '02_better_processed_data2.0.csv' – The processed dataset containing context, input, and output fields used for fine-tuning.
  3) Demo Video: '03_DemoVideo' - demo video of the GUI for Real-time Story Generator with Context Memory
  4) Evaluation Outputs:
     a '05a_eval_results_EvaluationMetrics.csv' – Contains automatic evaluation metrics (BLEU, ROUGE, BERTScore, Perplexity) for validation outputs.
     b '05b_eval_results_ManualMetrics.csv' – Contains manual evaluations (context relevance, coherence, creativity, fluency) scored on a 5-point Likert scale.
  5) Presentation slides: '06_Presentation_Slides' - slides presented during the class presentation.
- Environment Setup : Google Colab
  1) Mount Google Drive: Ensure the model and dataset are stored in Google Drive for persistence across Colab sessions. Mount drive in the notebook: from google.colab import drive drive.mount('/content/drive')
  2) Install Required Packages: Use the following commands to ensure compatibility: !pip install -q transformers==4.37.0 accelerate==0.22.0 peft==0.7.0 bitsandbytes==0.42.0 triton==2.0.0 datasets evaluate
  3) Login to Hugging Face: Authenticate to access gated models: from huggingface_hub import login login(token="your_huggingface_token")
- Running the notebook
  1) Data Preprocessing: Loads the dataset, applies TF-IDF and KMeans clustering for stratified sampling across short, medium, and long narratives.
  2) Model Loading: Loads the Mistral-7B-Instruct model with 4-bit quantization and LoRA adapters via QLoRA.
  3) Fine-tuning: Fine-tunes the model for 100 steps using SFTTrainer with batch size = 2 and gradient accumulation = 4.
  4) Evaluation: Evaluates generated outputs using automatic metrics (BLEU, ROUGE, BERTScore, Perplexity). Conducts manual assessments across key quality dimensions.
  5) Saving Reloading: Saves the fine-tuned model to Google Drive for reuse: model.save_pretrained("/content/drive/MyDrive/finetuned_mistral_short") tokenizer.save_pretrained("/content/drive/MyDrive/finetuned_mistral_short")
- Navigating the Results
  1) Loss Curve: Check the training loss graph toward the end of the notebook to assess training stability.
  2) Evaluation Metrics: Refer to 06a_eval_results_EvaluationMetrics.csv for detailed scores across automatic metrics. 06b_eval_results_ManualMetrics.csv contains human-assigned scores on context relevance, coherence, creativity, and fluency.
  3) Sample Generations: Review the generated outputs and reference texts in Sections 6.3 and 6.4 of the notebook.

References

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[6] Kaggle. Interactive Story Generation with BLOOMZ. https://www.kaggle.com/code/gkaran07/interactive-story-generation-with-bloomz.